Advanced Programming Practice · 21CSC203P

Unit 1 · Session 9 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 9 : SLO – 1 | Functional Programming Concepts

Activity (Solutions)

• 1) Pure function with Function<T,R>: java.util.function.Function<Integer,Integer> square = x -> x * x; System.out.println(square.apply(6)); // 36 — pure: same input always gives same output, no side effects.

• 2) Higher-order function = a function that takes a function as an argument or returns one. Java example: static int applyTwice(IntUnaryOperator f, int v){ return f.applyAsInt(f.applyAsInt(v)); } applyTwice(x -> x + 1, 5) → 7. Also list.forEach(System.out::println).

• 3) Immutability with final: final int x = 10; // x = 20; // compile error — reassignment forbidden. final List<String> names = List.of("a","b"); // names.add("c") throws UnsupportedOperationException — state cannot change after creation, a core functional principle.